Homework 4, due on 4/1/2026

1. The bandgap energy in a semiconductor is usually a slight function of temperature. In some cases, the bandgap energy versus temperature can be modeled by:

$$E_g = E_g(0) - \frac{\alpha T^2}{(\beta + T)}$$

where $E_g(0)$ is the value of the bandgap energy at T=0 K. please find the α and β value for GaAs from the literature. Plot Eg versus T over the range $0 \leq T \leq 600K$. In particular, note the value at T = 300 K. (10pts)

2. Suppose the band structure is the non-parabolic band as below, what is the effective mass at k=0? (10pts)

$$E(k)\left[1+\alpha E(k)\right]=\frac{\hbar^2 k^2}{2m^*(0)}$$

$E(\omega) + \alpha E^2(\omega) = \omega \quad E(\omega) = 0$

$\frac{\partial E}{\partial k} + 2\alpha E \cdot \frac{\partial E}{\partial k} = \frac{\hbar^2 k}{m(\omega)} \quad \frac{\partial E}{\partial k}\Big|_{k=0} = 0$

$\frac{\partial^2 E}{\partial k^2} + 2\alpha\left(\frac{\partial E}{\partial k}\right)^2 + 2\alpha E \cdot \frac{\partial^2 E}{\partial k^2} = \frac{\hbar^2}{m(\omega)} \quad \frac{\partial^2 E}{\partial k^2} = \frac{\hbar^2}{m(\omega)}$

$m^*(\omega) = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}} = m(\omega)$

3. Band structure calculation.
   a) Derive the equation below shown in the slides. (10pts)

$$\frac{-(\alpha^2+\beta^2)}{2\alpha\beta}(\sin\alpha a)(\sin\beta b) + (\cos\alpha a)(\cos\beta b) = \cos k(a+b)$$

   b) Assume a=0.45nm, b= 0.1nm, V_0=4eV, please plot the E vs k for the first 3 bands from k=–π/(a+b) to k=π/(a+b). (10pts)
   c) Please show the effective mass at k=0 for the first 3 band bands. (10pts)
   d) If you increase V_0, how the effective mass at k=0 changes for the first 3 bands? (5pts)
   e) If you increase b, how the effective mass at k=0 changes for the first 3 bands? (5pts)

4. Two possible conduction bands are shown in the E versus k diagram given in Figure 1. State which band will result in the heavier electron effective mass; state why. (10ps)

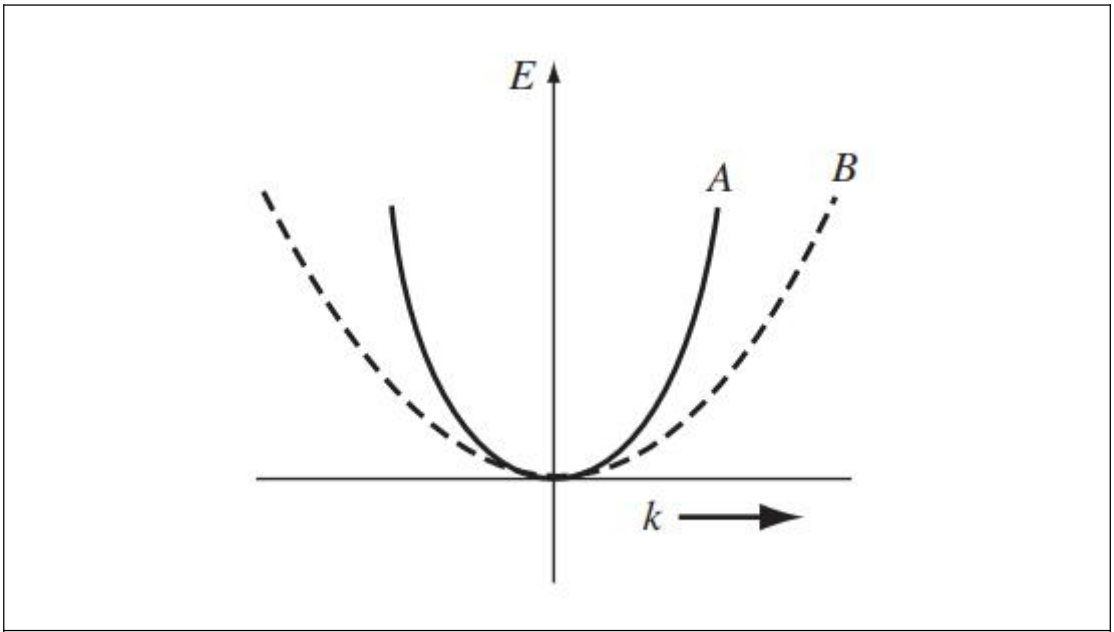


Figure 1. conduction band for problem 4.

$a = \frac{dv}{dt} = \frac{1}{\hbar}\frac{\partial^2 E}{\partial k}\frac{1}{\partial t}$ $a = \frac{F}{m^*} = \frac{F}{\hbar^2}\frac{\partial^2 E}{\partial k^2}$

$\Rightarrow \frac{F}{\hbar^2}\frac{\partial^2 E}{\partial k^2} = \frac{1}{\hbar}\frac{\partial^2 E}{\partial k \partial t}$

$F = \hbar \cdot \frac{\partial k}{\partial t}$

5. In the class, we show that the velocity of the electron is $v(k) = \frac{1}{\hbar}\frac{\partial E(k)}{\partial k}$, please prove $\frac{\hbar dk}{dt} = F$, where F is the external force. (10pts)
6. In the so-called tight binding method for computing band structures, the E(k) relation for one-dimensional lattice is given by

$$\mathrm{E(k) = A - Bcos(ka)}$$

where A and B are constants, and a is the lattice spacing. Using this band structure for one-dimensional electrons, answer the following questions. Assume A=B=1, a = 5e-10.

a) Plot the E(k) relation for $-\pi/a \leq k \leq \pi/a$ (5pts)
b) Plot the velocity, v(k), for $-\pi/a \leq k \leq \pi/a$ (5pts)
c) Assuming that the electrical field is $-\epsilon_0$, how long would it take an electron to travel from k=0 to k =π/2a? (5pts)
d) Given the answer T0 from part c, how far would the electron go in this time? (5pts)

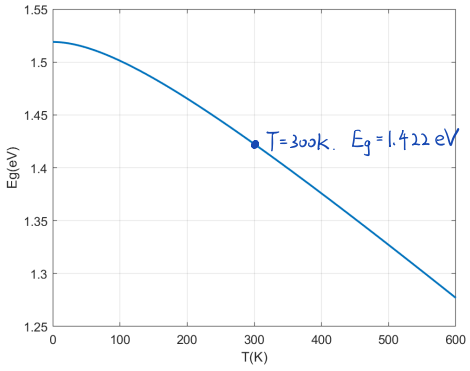
$\frac{\hbar dk}{dt} = -\epsilon_0 \cdot e$

$\Rightarrow t = \frac{\hbar\pi}{2a}\cdot\frac{1}{\epsilon_0 e}$

1. From literature, $\alpha = 5.405 \times 10^{-4}\ eV/k$, $\beta = 204\ k$, $E_g(0) = 1.519\ eV$

$$E_g = 1.519 - \frac{5.405 \times 10^{-4} T^2}{204 + T}$$

take $T = 300k$. $E_g = 1.422\ eV$



2. $E(k)[1 + \alpha E(k)] = \frac{\hbar^2 k^2}{2m_0}$

$$E(0) + \alpha \cdot E^2(0) = 0$$

$$\Rightarrow E(0) = 0$$

$$\frac{\partial E(k)}{\partial k} + 2\alpha \cdot E(k)\frac{\partial E(k)}{\partial k} = \frac{\hbar^2 k}{m_0}$$

$$\Rightarrow \left.\frac{\partial E(k)}{\partial k}\right|_{k=0} = 0$$

$$\frac{\partial^2 E(k)}{\partial k^2} + 2\alpha\left[\left(\frac{\partial E(k)}{\partial k}\right)^2 + E(k)\cdot\frac{\partial^2 E(k)}{\partial k^2}\right] = \frac{\hbar^2}{m_0}$$

$$\Rightarrow \left.\frac{\partial^2 E(k)}{\partial k^2}\right|_{k=0} = \frac{\hbar^2}{m_0}$$

$\Rightarrow$ effective mass at $k=0$: $m^*(0) = \frac{\hbar^2}{\frac{\partial^2 E(k)}{\partial k^2}} = m_0$

3. a) Krony - Penney model



while $0 \le x \le a$, $-\frac{\hbar^2}{2m}\cdot\varphi_a'' = E\cdot\varphi_a$

$\Rightarrow \varphi_a(x) = A_a\cdot\sin\alpha x + B_a\cos\alpha x \qquad \alpha^2 = \frac{2mE}{\hbar^2}$

while $-b \le x \le 0$, $-\frac{\hbar^2}{2m}\cdot\varphi_b'' + V_0\cdot\varphi_b = E\cdot\varphi_b$

$\Rightarrow \varphi_b(x) = A_b\cdot\sin\beta x + B_b\cdot\cos\beta x \quad \beta^2 = \frac{2m(E-V_0)}{\hbar^2}$

from Bloch Theorem, $\varphi_a(a^+) = \varphi_b(-b)\,e^{ikp}$, $\varphi_a'(a^+) = \varphi_b'(-b)\,e^{ikp}$, where $p = a+b$

$$\Rightarrow \begin{cases} \varphi_a(0) = \varphi_b(0) \\ \varphi_a'(0) = \varphi_b'(0) \\ \varphi_a(a^-) = \varphi_b(-b)\,e^{ikp} \\ \varphi_a'(a^-) = \varphi_b'(-b)\,e^{ikp} \end{cases} \Longrightarrow \begin{cases} B_a = B_b \\ A_a\alpha = A_b\beta \\ A_a\sin\alpha a + B_a\cos\alpha a = (-A_b\sin\beta b + B_b\cos\beta b)\,e^{ikp} \\ A_a\alpha\cos\alpha a - B_a\alpha\sin\alpha a = (A_b\beta\cos\beta b + B_b\beta\sin\beta b)\,e^{ikp} \end{cases}$$

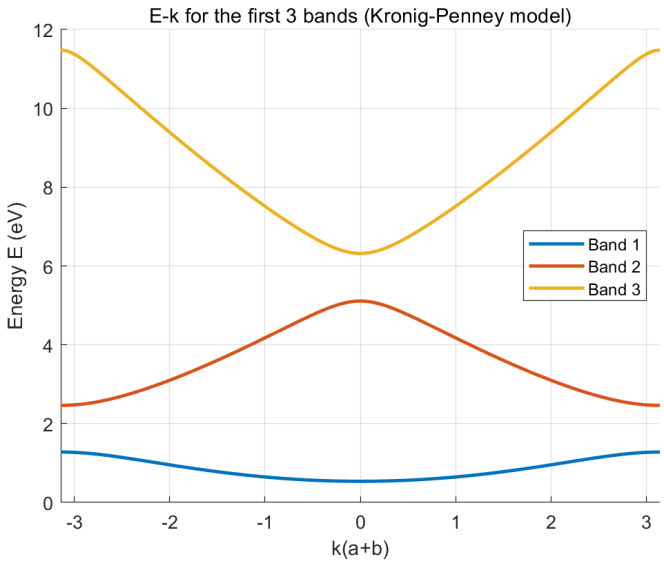
$$\Rightarrow \underbrace{\begin{bmatrix} 0 & 0 & 1 & -1 \\ \alpha & -\beta & 0 & 0 \\ \sin\alpha a & e^{ikp}\sin\beta b & \cos\alpha a & -e^{ikp}\cos\beta b \\ \alpha\cos\alpha a & -e^{ikp}\beta\cos\beta b & -\alpha\sin\alpha a & -e^{ikp}\beta\sin\beta b \end{bmatrix}}_{\tilde{M}(E,k)} \begin{bmatrix} A_a \\ A_b \\ B_a \\ B_b \end{bmatrix} = 0$$

To solve A_a, A_b, B_a, B_b, we have $\det(\tilde{M}(E,k)) = 0$

$$\Rightarrow \det\begin{pmatrix} 0 & 0 & 1 & -1 \\ \alpha & -\beta & 0 & 0 \\ \sin\alpha a & e^{ikp}\sin\beta b & \cos\alpha a & -e^{ikp}\cos\beta b \\ \alpha\cos\alpha a & -e^{ikp}\beta\cos\beta b & -\alpha\sin\alpha a & -e^{ikp}\beta\sin\beta b \end{pmatrix} = \begin{vmatrix} \alpha & -\beta & 0 \\ \sin\alpha a & e^{ikp}\sin\beta b & -e^{ikp}\cos\beta b \\ \alpha\cos\alpha a & -e^{ikp}\beta\cos\beta b & -e^{ikp}\beta\sin\beta b \end{vmatrix} + \begin{vmatrix} \alpha & -\beta & 0 \\ \sin\alpha a & e^{ikp}\sin\beta b & \cos\alpha a \\ \alpha\cos\alpha a & -e^{ikp}\beta\cos\beta b & -\alpha\sin\alpha a \end{vmatrix} = 0$$

$$\Rightarrow -\frac{\alpha^2+\beta^2}{2\alpha\beta}\sin\alpha a\cdot\sin\beta b + \cos\alpha a\cdot\cos\beta b = \cos k(a+b)$$

b)



c) at $k=0$. $-\frac{\alpha^2+\beta^2}{2\alpha\beta}\sin\alpha a \cdot \sin\beta b + \cos\alpha a \cdot \cos\beta b = \cos k(a+b)$

$\alpha = \frac{\sqrt{2mE}}{\hbar}$, $\beta = \frac{\sqrt{2m(E-V_0)}}{\hbar}$

let $F(E,k) = -\frac{\alpha^2+\beta^2}{2\alpha\beta}\sin\alpha a \cdot \sin\beta b + \cos\alpha a \cdot \cos\beta b - \cos k(a+b) = 0$

using matlab to solve $\frac{\partial F}{\partial k}\Big|_{k=0} = \frac{\partial^2 F}{\partial k^2}\Big|_{k=0} = 0$ numerically.

effective mass: $m^* = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}}$

we have: Band 1: $m^* = 1.027 \times 10^{-30}\ kg$

Band 2: $m^* = -6.214 \times 10^{-32}\ kg$

Band 3: $m^* = 5.443 \times 10^{-32}\ kg$.

d) If we increase V_0, the Band will be more flat, which is the difficulty for a electron jump to another energy band increases, thus curvature decreases.

$\Rightarrow \left|\frac{\partial^2 E}{\partial k^2}\right| \downarrow \Rightarrow |m^*| = \frac{\hbar^2}{\left|\frac{\partial^2 E}{\partial k^2}\right|} \uparrow$. $|m^*|$ for the first 3 band increased. m^* for Band 1 and 3 increased, for Band 2 decreased.

e) If we increase b, the Band will be more isolated the difficulty for a electron jump to another energy band increases, thus curvature decreases.

$\Rightarrow \left|\frac{\partial^2 E}{\partial k^2}\right| \downarrow \Rightarrow |m^*| = \frac{\hbar^2}{\left|\frac{\partial^2 E}{\partial k^2}\right|} \uparrow$. $|m^*|$ for the first 3 band increased. m^* for Band 1 and 3 increased, for Band 2 decreased.

4. We can see that the curvature of A is greater than B. $\frac{\partial^2 E_A}{\partial k^2} > \frac{\partial^2 E_B}{\partial k^2}$

Since $m^* = \frac{\hbar^2}{\frac{\partial^2 E}{\partial k^2}}$

we have effective mass : $m^*_B > m^*_A$

5. $v_{(k)} = \frac{1}{\hbar} \cdot \frac{\partial E(k)}{\partial k}$

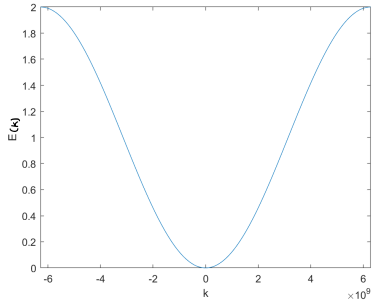
$a = \frac{\partial v}{\partial t} = \frac{1}{\hbar} \frac{\partial^2 E(k)}{\partial k \partial t}$

$$\begin{cases} \sum F = m^* a \\ m^* = \frac{\hbar^2}{\frac{\partial^2 E(k)}{\partial k^2}} \end{cases}$$

$\Rightarrow F = \frac{\hbar^2}{\frac{\partial^2 E(k)}{\partial k^2}} \cdot \frac{1}{\hbar} \cdot \frac{\partial^2 E(k)}{\partial k \partial t}$

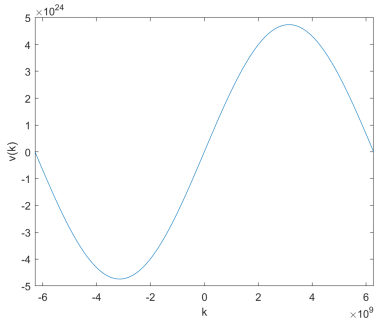
$\Rightarrow F = \hbar \cdot \frac{dk}{dt}$

6. a) $E(k) = 1 - \cos(k \times 5 \times 10^{-10})$



b) $v_{(k)} = \frac{1}{\hbar} \cdot \frac{\partial E(k)}{\partial k}$

$v_{(k)} = \frac{a \cdot B}{\hbar} \cdot \sin(ka)$



c) $\hbar \cdot \frac{dk}{dt} = F$

$F = e \cdot \epsilon_0 \Rightarrow \hbar \frac{dk}{dt} = e \cdot \epsilon_0$

$dt = \frac{\hbar}{e \cdot \epsilon_0} \cdot dk$

Assume it would take t_0 for a electron to travel from $k=0$ to $k=\frac{\pi}{2a}$.

$\int_0^{t_0} dt = \frac{\hbar}{e \cdot \epsilon_0} \cdot \int_0^{\frac{\pi}{2a}} dk$

$t_0 = \frac{\pi \hbar}{2ae\epsilon_0}$

$\Rightarrow t_0 = \frac{2.07 \times 10^{-6}}{\epsilon_0}$

d) $v(k) = \frac{1}{\hbar} \cdot \frac{\partial E(k)}{\partial k}$

$\Rightarrow v(k) = \frac{a}{\hbar} \cdot \sin(ka)$

$\chi = \int_0^{T_0} v(k) \, dt$

Since $dt = \frac{\hbar}{e \cdot \epsilon_0} dk$

$\chi = \frac{a}{e \cdot \epsilon_0} \cdot \int_0^{\frac{\pi}{2a}} \sin(ka) \, dk$

$\chi = \frac{1}{e \cdot \epsilon_0}$

$\Rightarrow \chi = \frac{1}{1.6 \times 10^{-9} \epsilon_0}$